Mechanism design

Some references

See Myerson (1981), Mas-Colell et al. (1995), Krishna (2010), and Menezes and Monteiro (2005).

Mechanism design in IPVM

- Goal. What is the **best** *way* to transfer the object?
- *best?* Objective function. Among possibilities:
  - seller's expected revenue.
  - aggregate surplus (buyers surplus).
- *way?* Mechanisms. Problem: too many.
- How to formalize the notion of mechanism so as to cover as many as possible transaction forms while still keeping the optimization problem tractable?
- How to solve the optimization program?

Definitions

Definition 1 (Environment) An environment is an $E = [O, \{(X_i, \mathcal{X}_i, \mu_i), V_i\}_{i=1}^{I}]$ where O is a set of outcomes, $(X_i, \mathcal{X}_i, \mu_i)$ is the space of agent i's private information, and $V_i : X \times O \to \mathbb{R}$ is i's payoffs given a realization of information and an outcome.

Definition 2 (Mechanism) Given E, a mechanism (or game form) is an $M = \{\{S_i\}_{i=1}^{I}, g : \prod_{i=1}^{I} S_i \to O\}$ where S_i is player i's set of available messages or signals, and g is a function, termed the outcome function, that assigns to each message profile an outcome.

Environment

environment

$\mathbb{R}^I$

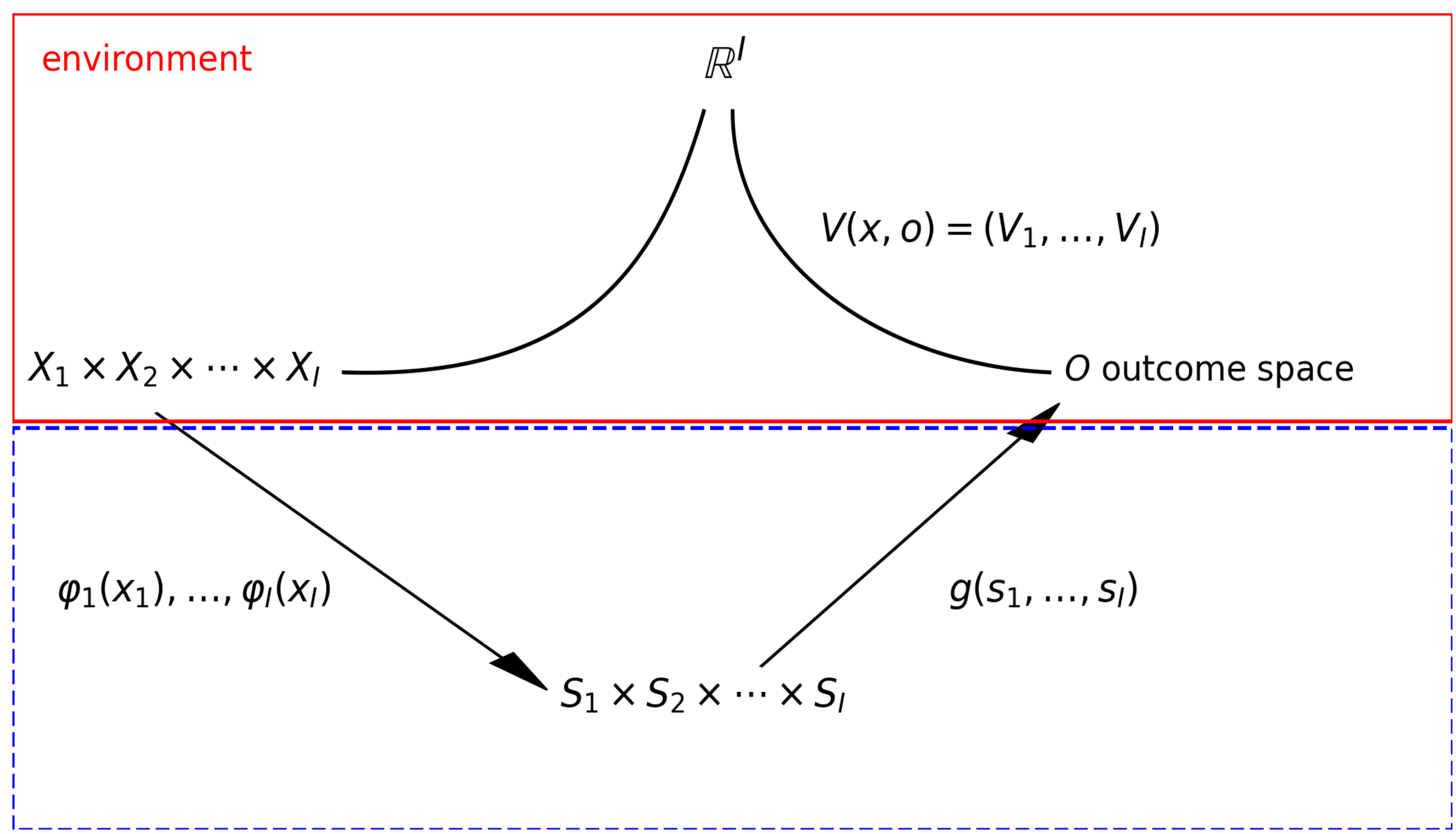
$V(x, o) = (V_1, \ldots, V_I)$

$o \in O, x \in X_1 \times X_I$

$X_1 \times X_2 \times \cdots \times X_I$

O outcome space

Environment and mechanism



- Given an environment E, every mechanism M induces a Bayesian game $\Gamma_M = \{(X_i, \mathcal{X}_i, \mu_i), S_i, u_i\}_{i=1}^{I}$,
- Player i observes private information $x_i \in X_i$ and chooses an action $s_i \in S_i$.

  A strategy for player i is a function $\phi_i : X_i \to S_i$
- Given an action profile s, $g(s) \in O$ is the outcome of the game.
- Player i's payoff is $u_i(x, s) = V_i(x, g(s))$.

$$u_i(x_1, \dots, x_n, q_1, q_2, \dots, q_n, m_1, \dots, m_n) = x_i \cdot q_i - m_i.$$

Direct mechanism

Definition 3 (Direct mechanism) A mechanism $M = [\{S_i\}_{i=1}^I, g]$ is *direct* if $S_i = X_i$ for every i.

- *Bidders* observe their types and report x' individually and independently.
- Given a report profile $x' = (x'_1, \dots, x'_I)$, the function g determines the outcome $g(x') \in O$.

Definition 4 (Direct revelation mechanism) If the game Γ_M induced by a direct mechanism M has an equilibrium $\{\phi_i\}_{i=1}^I$ where for every i and every $x_i \in X_i$, $\phi(x_i) = x_i$, then M is a *direct revelation mechanism* (DRM).

- BNE versus DSE.

Example: application to the IPVM

- $O = \{(q, m) : q \in [0,1]^I, \sum_{i=1}^I q_i \leq 1, m \in \mathbb{R}^I\}$
- A mechanism $M = [\{S_i\}_{i=1}^I,\ g : \prod_{i=1}^I S_i \to O]$:
- A signaling space S_i per agent, $s_i \in S_i$, $s = (s_1, s_2, \dots, s_I)$.
- $g(s) = (q(s), t(s))$,
  - $q(s) = (q_1(s), \dots, q_I(s))$, $\sum_{i=1}^I q(s) \leq 1$,
  - $t(s) = (t_1(s), \dots, t_I(s))$.

Example FPA: mechanisms that is not direct, $x_i \in [0, 1]$

- $S_i = \mathbb{R}$, $b_i \in \mathbb{R}$ is i's bid.
- $g(b) = g(b_1, \ldots, b_I) = \{q_i(b), t_i(b)\}_{i=1}^{I}$
- $q_i(b) = \begin{cases} 1 & \text{if } b_i > b_{-i}^{(1)} \\ 0 & \text{if } b_i < b_{-i}^{(1)} \\ . & \text{if } b_i = b_{-i}^{(1)} \end{cases}$
- $t_i(b) = b_i q_i(b)$.

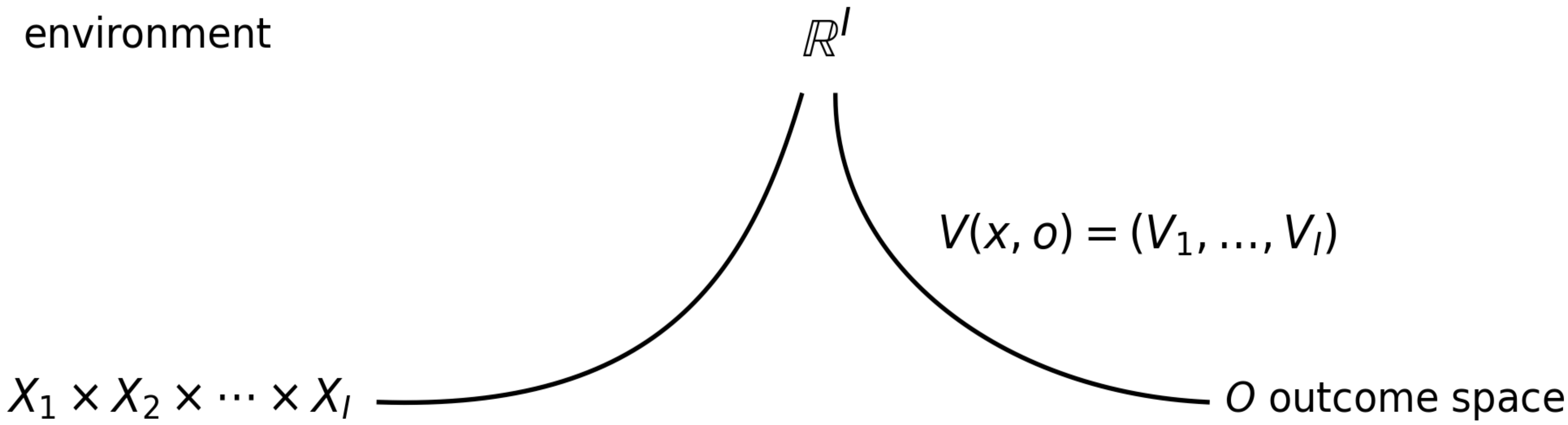
Example SPA: (it can be seen as a direct mechanism)

- $S_i = X_i = [0,1]$, $b_i \in X_i = [0,1]$ is i's report (and bid).
- $q_i(b)$ as in the FPA.
- $t_i(b) = \begin{cases} b_{-i}^{(1)} & \text{if } b_{-i}^{(1)} < b_i \\ 0 & \text{if } b_{-i}^{(1)} > b_i \\ . & \text{if } b_{-i}^{(1)} = b_i \end{cases}$

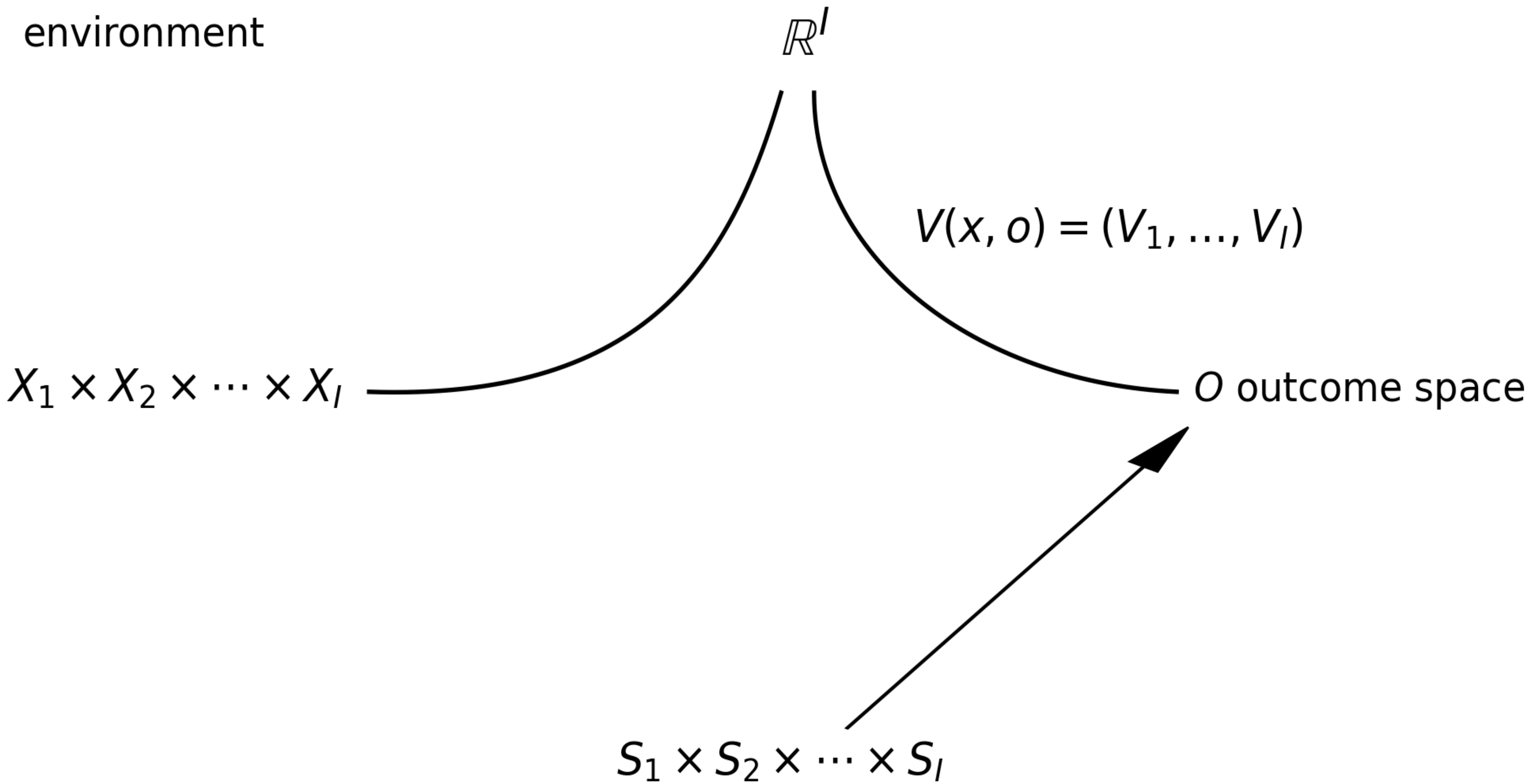
Revelation principle

Theorem 1 (Revelation principle) Let ϕ be an equilibrium of Γ_M, the game induced by a mechanism M. Then there exists an equilibrium ϕ' of (the game induced by) a direct mechanism g' such that the ϕ' and ϕ generate the same outcome and for every $\forall i \ \wedge x_i \in X_i, \phi_i'(x_i) = x_i$.

Proof 1

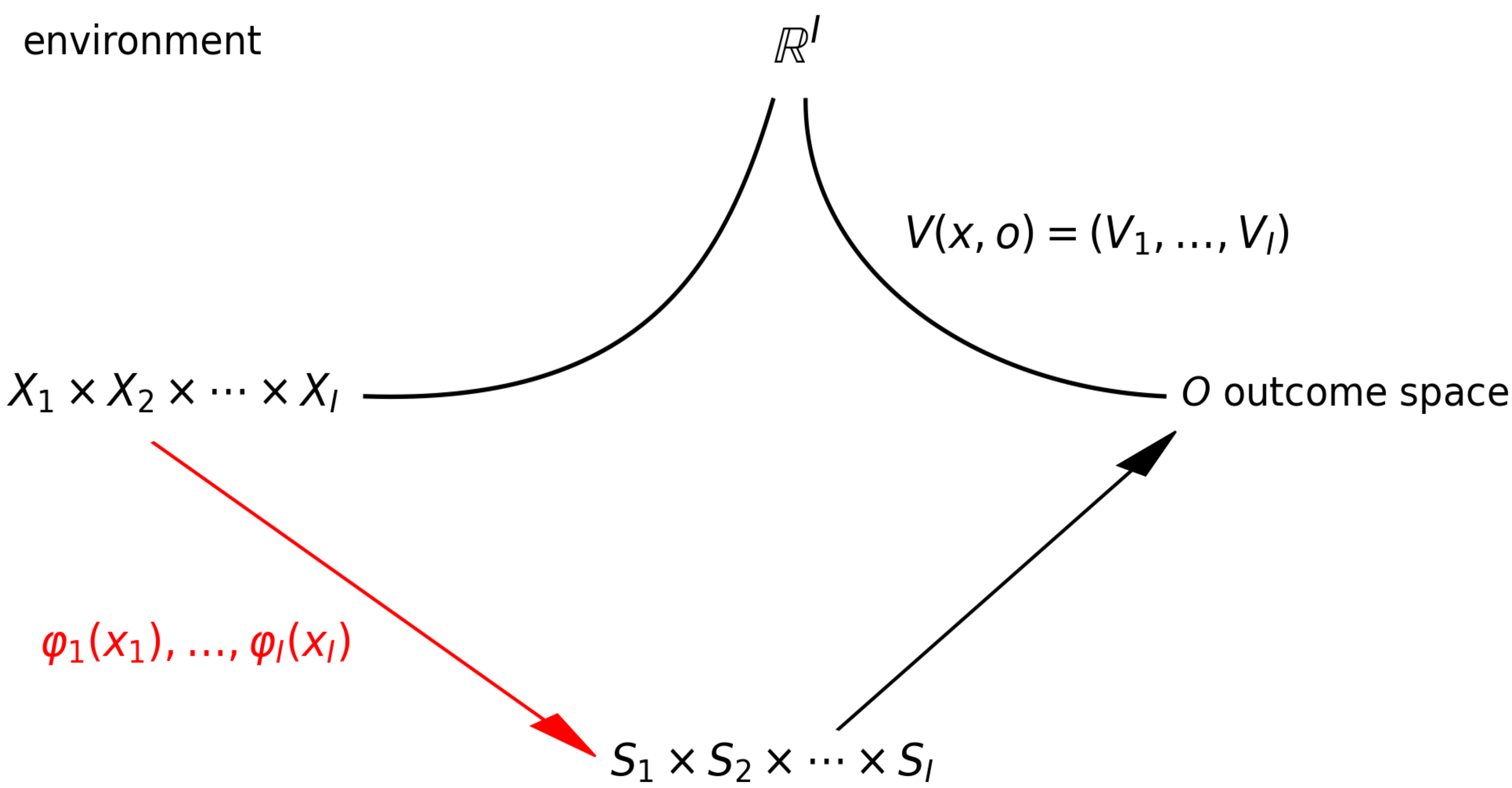

environment
$\mathbb{R}^I$
$V(x, o) = (V_1, \ldots, V_I)$
$X_1 \times X_2 \times \cdots \times X_I$
O outcome space

Proof 2



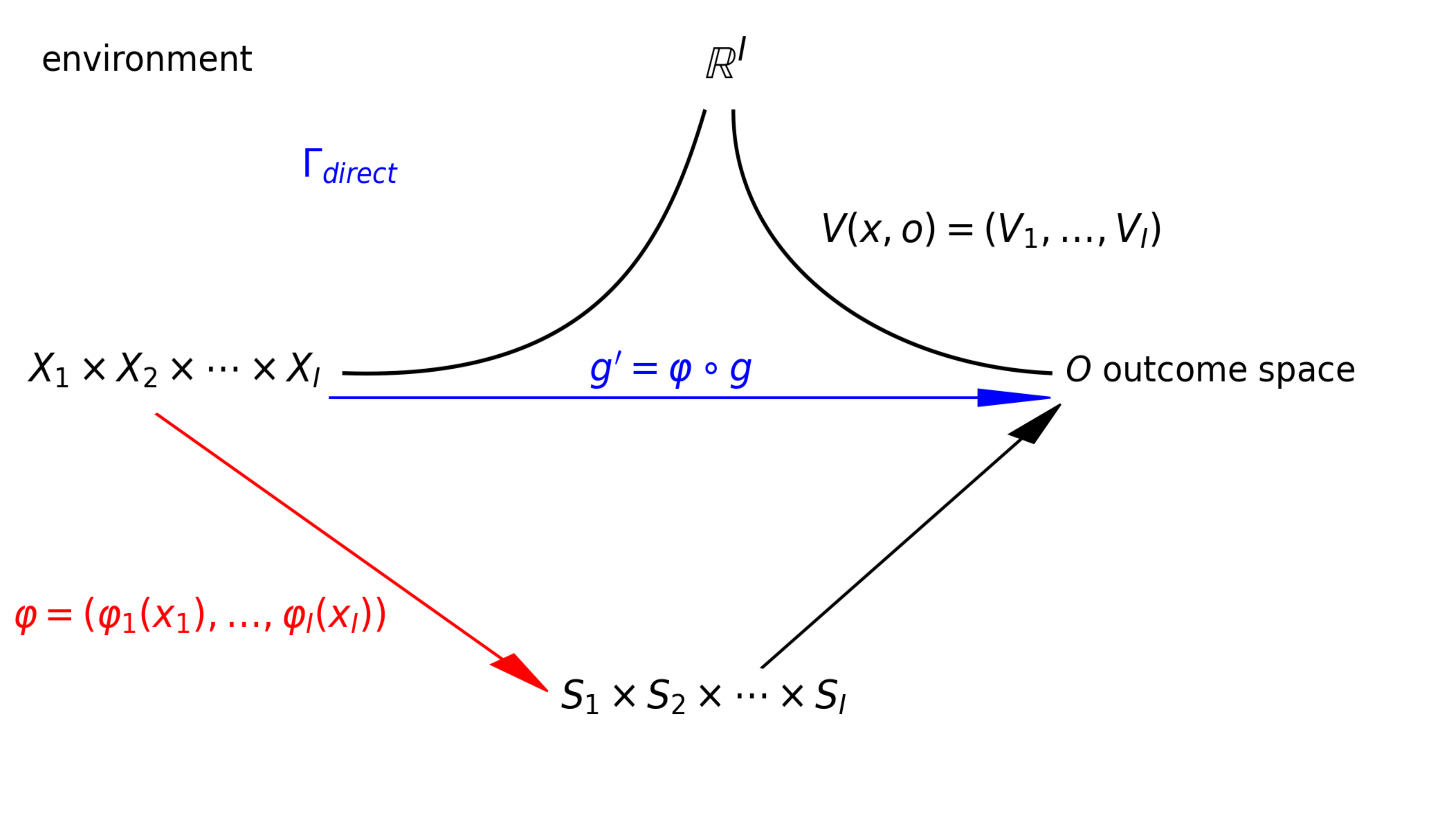
environment
$\mathbb{R}^I$
$V(x, o) = (V_1, \ldots, V_I)$
$X_1 \times X_2 \times \cdots \times X_I$
O outcome space
$S_1 \times S_2 \times \cdots \times S_I$

Proof 3

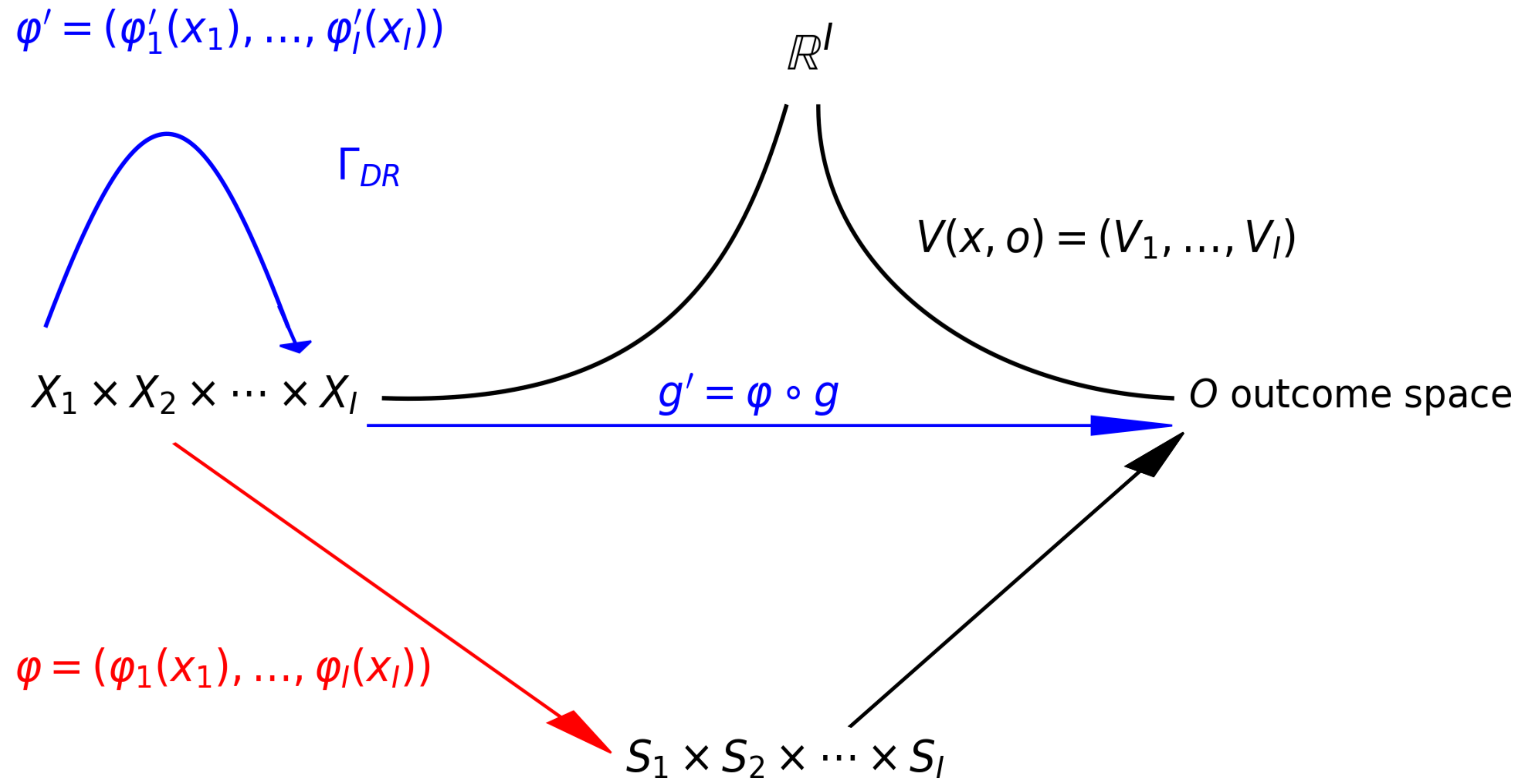


environment
$\mathbb{R}^I$
$V(x, o) = (V_1, \ldots, V_I)$
$X_1 \times X_2 \times \cdots \times X_I$
O outcome space
$\varphi_1(x_1), \ldots, \varphi_I(x_I)$
$S_1 \times S_2 \times \cdots \times S_I$

Proof 4


environment
$\mathbb{R}^I$
Γ_{direct}
$V(x, o) = (V_1, \dots, V_I)$
$X_1 \times X_2 \times \cdots \times X_I$
$g' = \varphi \circ g$
O outcome space
$\varphi = (\varphi_1(x_1), \dots, \varphi_I(x_I))$
$S_1 \times S_2 \times \cdots \times S_I$

Proof 5



$\varphi' = (\varphi'_1(x_1), \ldots, \varphi'_I(x_I))$
$\mathbb{R}^I$
Γ_{DR}
$V(x, o) = (V_1, \ldots, V_I)$
$X_1 \times X_2 \times \cdots \times X_I$
$g' = \varphi \circ g$
O outcome space
$\varphi = (\varphi_1(x_1), \ldots, \varphi_I(x_I))$
$S_1 \times S_2 \times \cdots \times S_I$

Proof (Sketch).

- Define DRM
- $(\forall i)\; S_i = X_i = [0, 1]$.
- $(\forall i, \forall x \in X = [0, 1]^I)$

  $q_i'(x_{-i}, x_i) = q_i(\phi_1(x_1), \cdots, \phi_I(x_I))$

  and

  $t_i'(x_{-i}, x_i) = t_i(\phi_{-i}(x_{-i}), \phi_i(x_i))$.
- Check that (q', t') generate the same outcome as (q, t) and that truth-telling is a BNE under (q', t') to complete the proof.

Modeling question

- Modeling question: model of mechanisms must be broad and tractable.
- Revelation Principle implies two simplifications
  1. We may restrict attention to direct mechanisms, message space is space of private information.
  2. We may farther restrict attention to direct mechanisms that have truthful revelation in equilibrium.
- It is clear how to use (1); how does one take advantage of (2)?
- Remark. Careful with use of Revelation Principle.

References

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